# CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

## Program 5 : Python Program for Missionaries and Cannibal Problem

### *Aim*

To develop a Python program that moves missionaries and cannibals across the river safely without the cannibals outnumbering the missionaries.

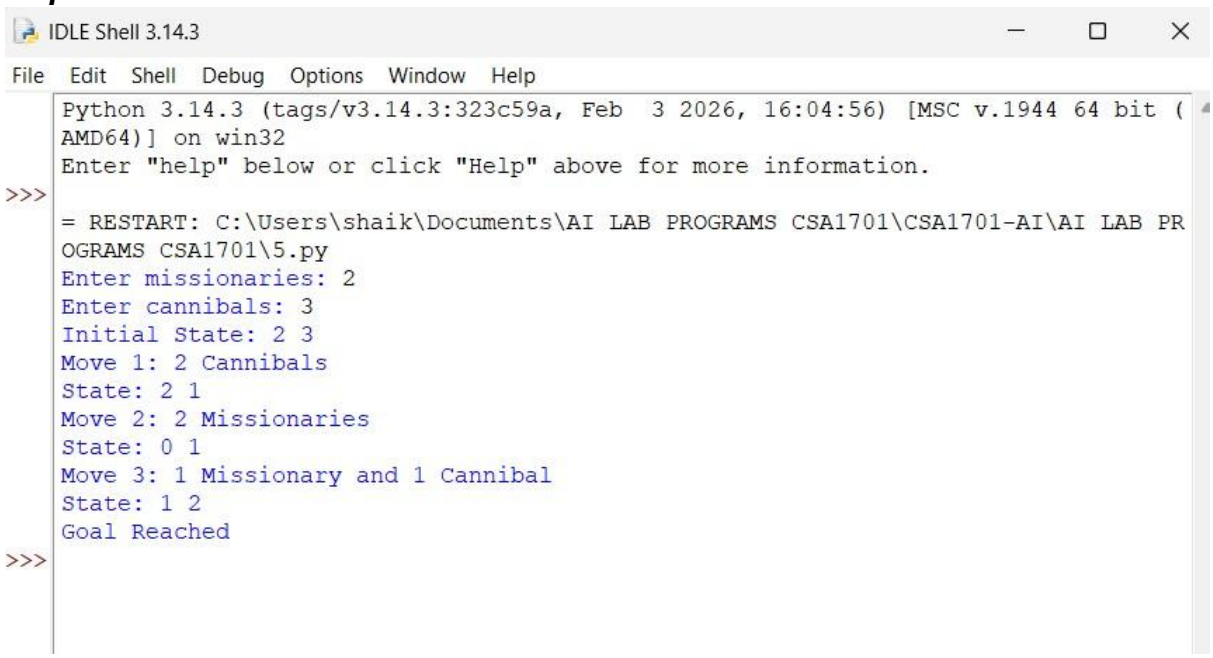
### *Algorithm*

- Read the number of missionaries and cannibals on the starting bank.
- Display the initial state.
- Move 1: take 2 cannibals across the river and update the state.
- Move 2: take 2 missionaries across the river and update the state.
- Move 3: bring 1 missionary and 1 cannibal back and update the state.
- Display the state after every move.
- Print "Goal Reached" when all the moves are completed.

***Program***

```
m = int(input("Enter missionaries: ")) c
= int(input("Enter cannibals: "))
print("Initial State:", m, c)
print("Move 1: 2 Cannibals")
c -= 2 print("State:", m, c)
print("Move 2: 2 Missionaries")
m -= 2 print("State:", m, c)
print("Move 3: 1 Missionary and 1
Cannibal") m += 1 c += 1 print("State:", m,
c) print("Goal Reached")
```

***Output***




***RESULT***

Thus, the Missionaries and Cannibal Problem was solved successfully using Python.